

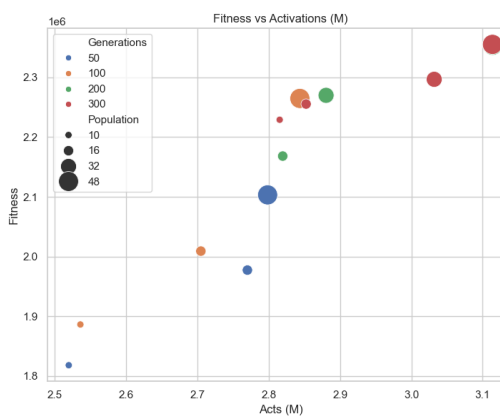
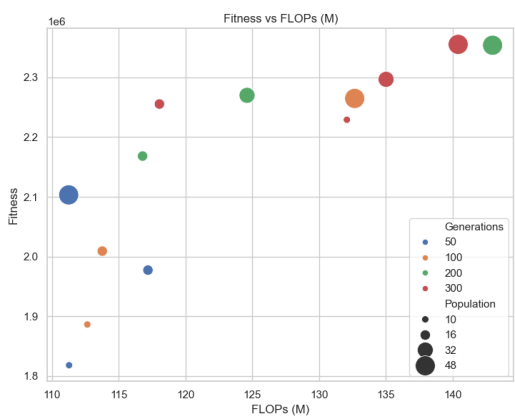
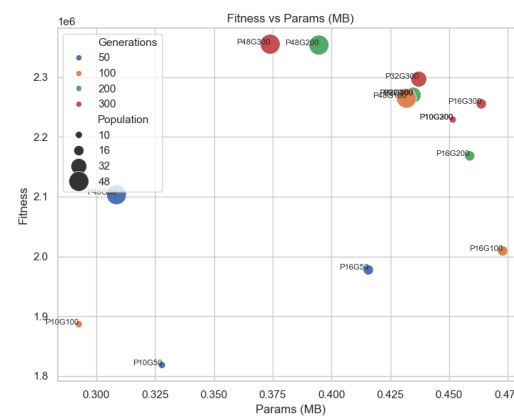
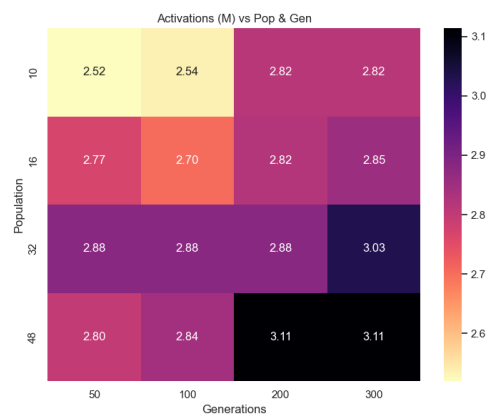
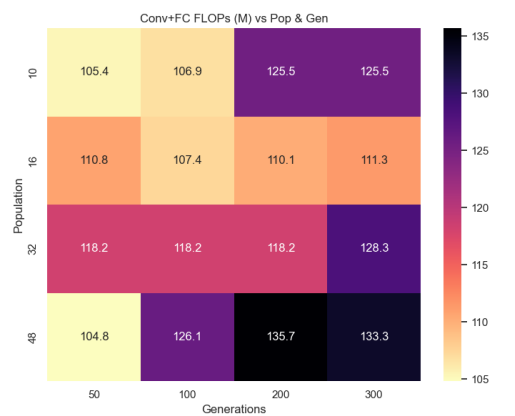
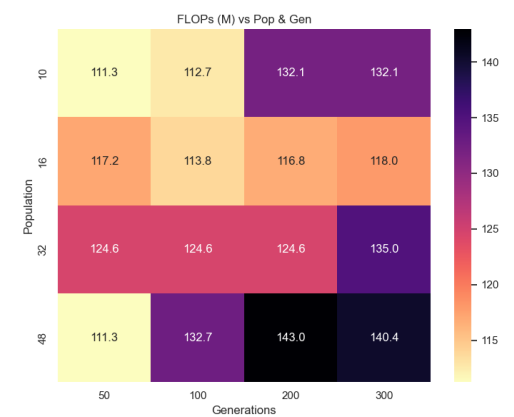
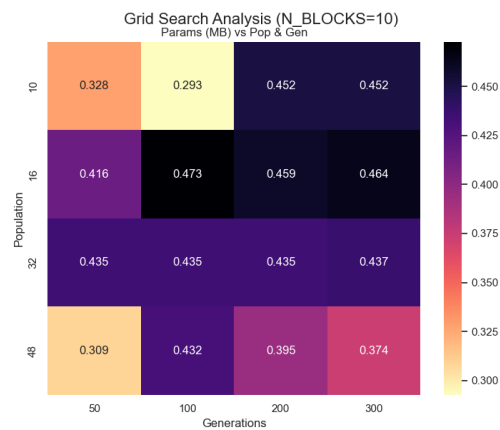
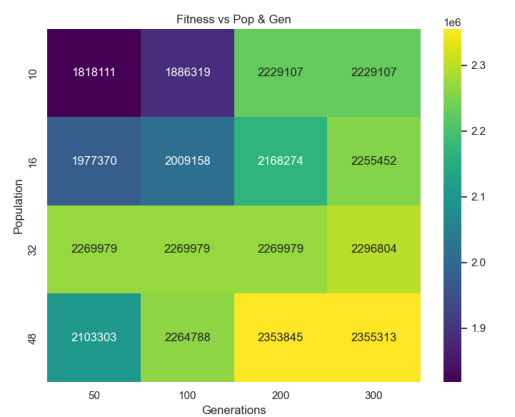
Grid Search Analysis Report

Comparison of Global EA vs Layer EA

Generated by NAS_LEGO

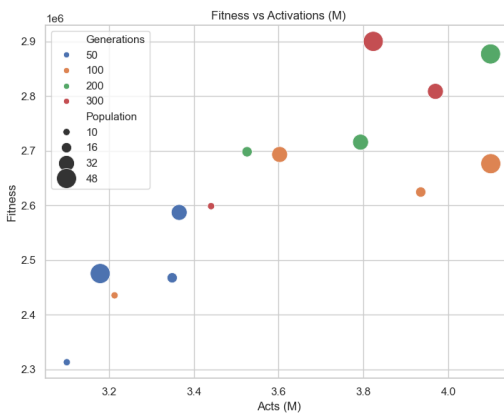
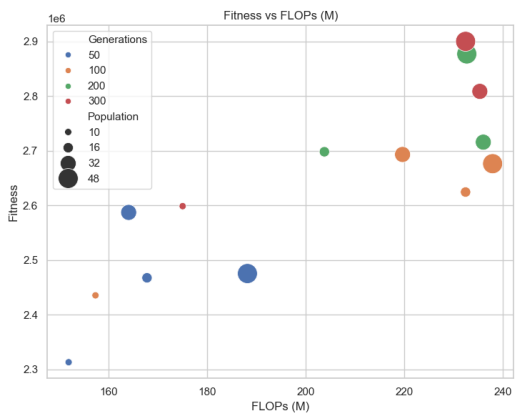
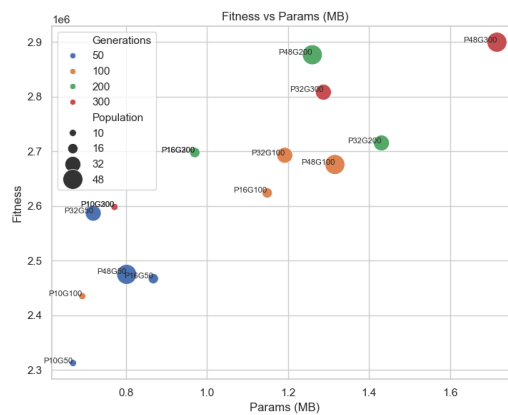
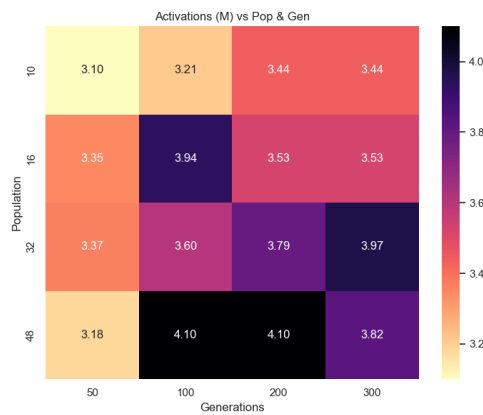
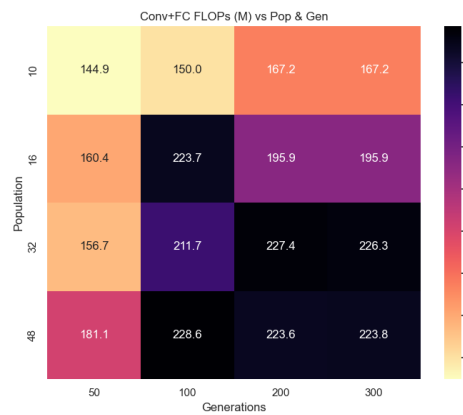
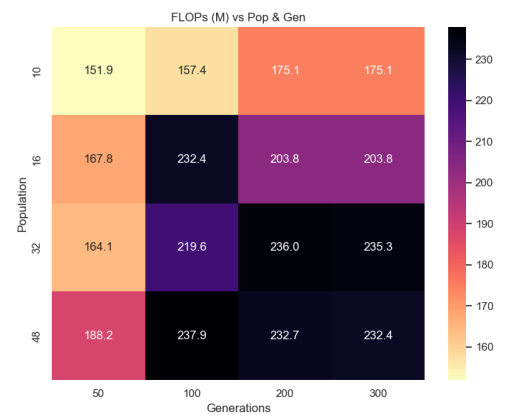
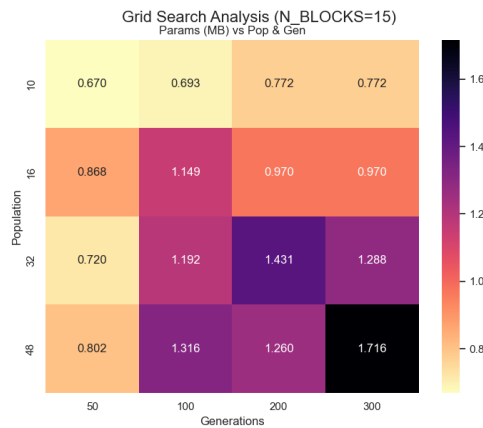
Section 1: Global Evolutionary Algorithm

Global EA - 10 Blocks



	Fitness	Params (MB)	FLOPs (M)	Conv+FC FLOPs (M)	Acts (M)
Fitness	1.0	0.5	0.8	0.8	0.91
Params (MB)	0.5	1.0	0.3	0.29	0.35
FLOPs (M)	0.8	0.3	1.0	1.0	0.83
Conv+FC FLOPs (M)	0.8	0.29	1.0	1.0	0.81
Acts (M)	0.91	0.35	0.83	0.81	1.0

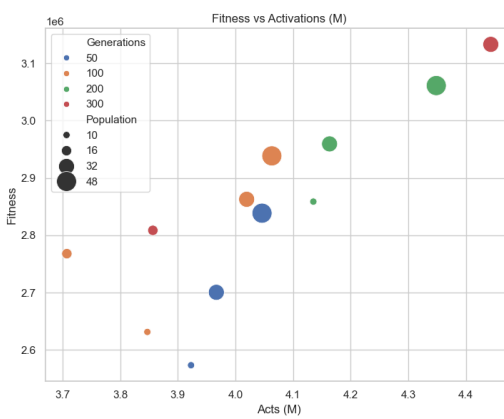
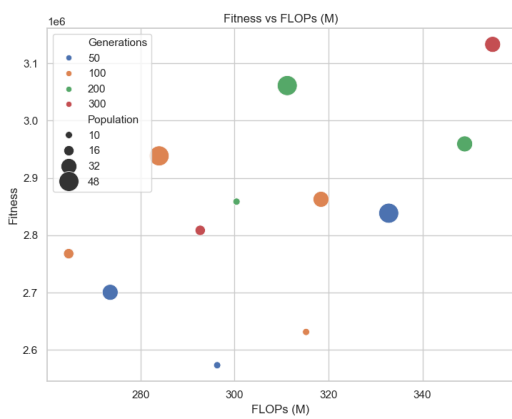
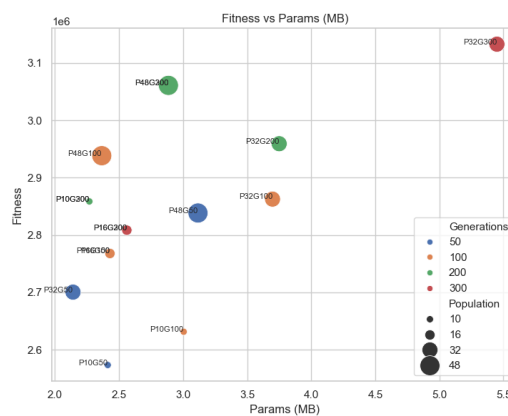
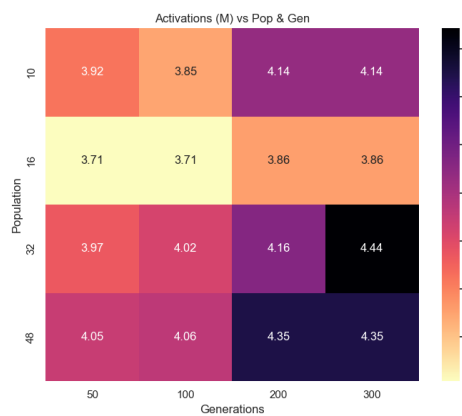
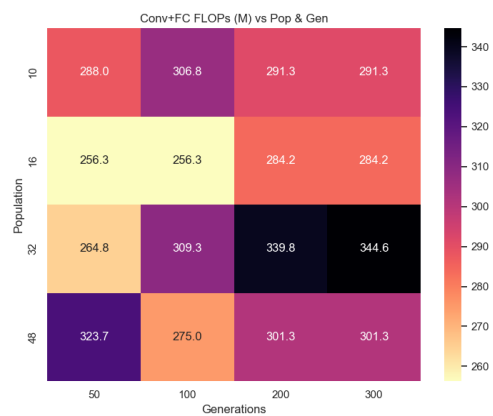
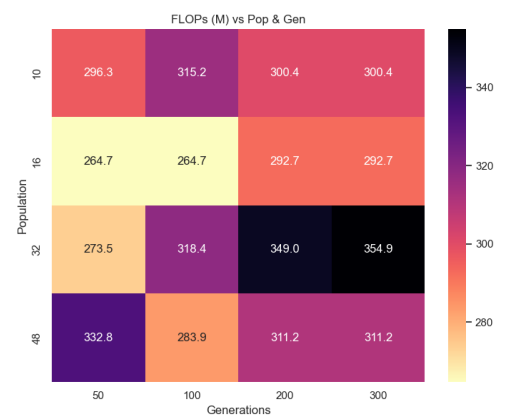
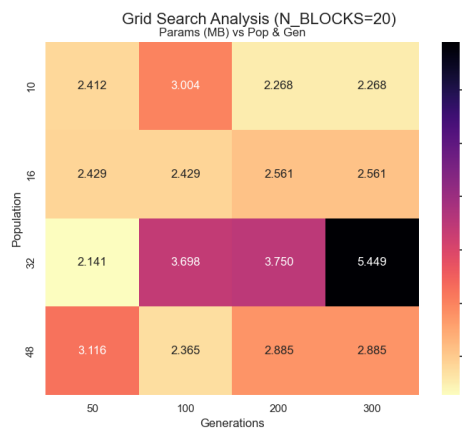
Global EA - 15 Blocks



Correlation Matrix

	Fitness	Params (MB)	FLOPs (M)	Conv+FC FLOPs (M)	Acts (M)
Fitness	1.0	0.82	0.83	0.82	0.82
Params (MB)	0.82	1.0	0.9	0.9	0.82
FLOPs (M)	0.83	0.9	1.0	1.0	0.91
Conv+FC FLOPs (M)	0.82	0.9	1.0	1.0	0.91
Acts (M)	0.82	0.82	0.91	0.91	1.0

Global EA - 20 Blocks



Correlation Matrix

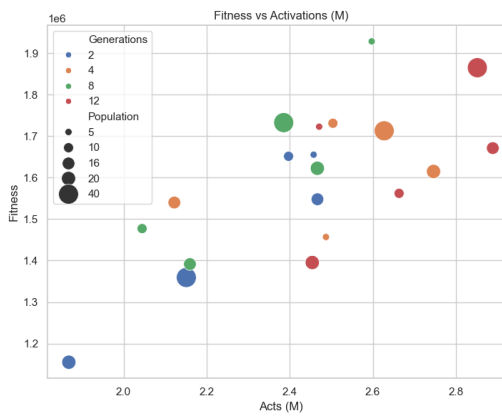
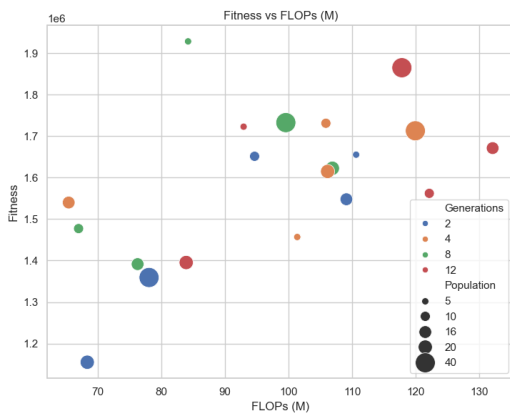
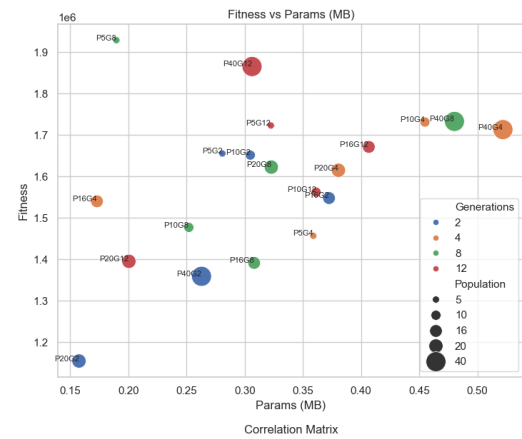
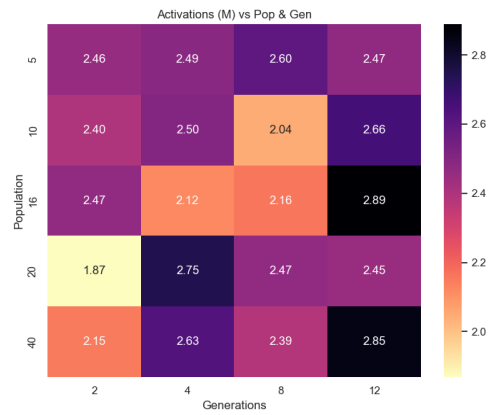
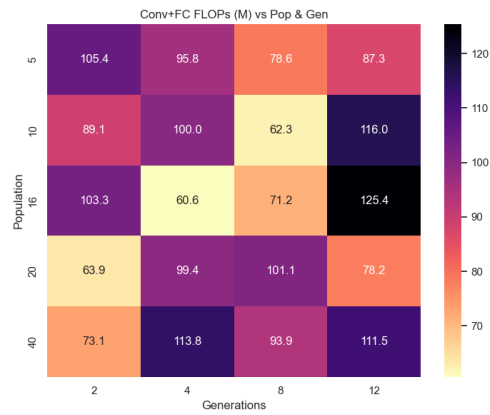
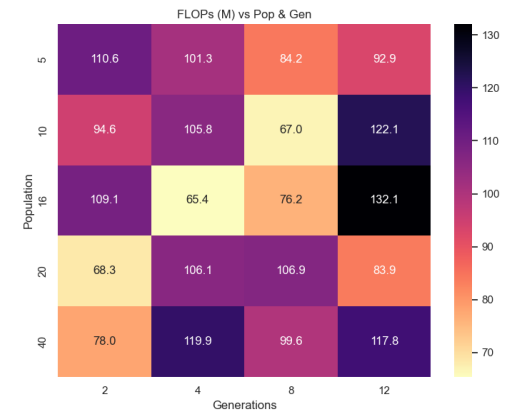
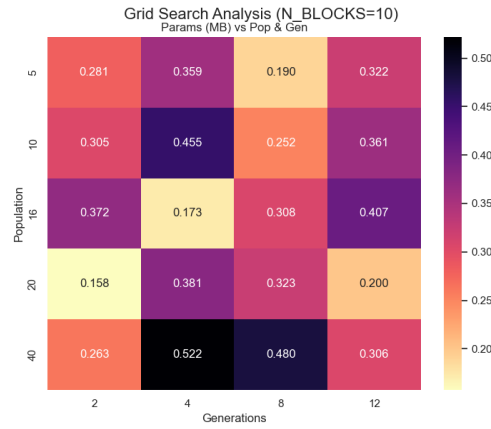
	Fitness	Params (MB)	FLOPs (M)	Conv+FC FLOPs (M)	Acts (M)
Fitness	1.0	0.57	0.52	0.51	0.82
Params (MB)	0.57	1.0	0.82	0.82	0.54
FLOPs (M)	0.52	0.82	1.0	1.0	0.69
Conv+FC FLOPs (M)	0.51	0.82	1.0	1.0	0.67
Acts (M)	0.82	0.54	0.69	0.67	1.0

Global EA - 5 Blocks



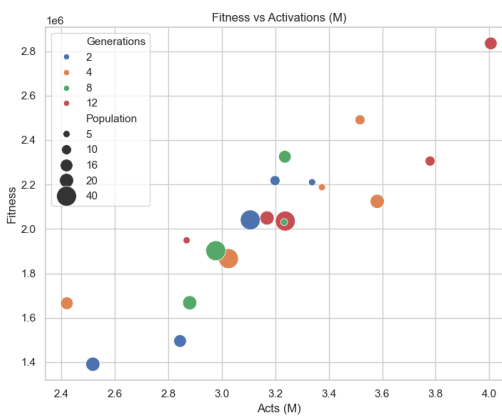
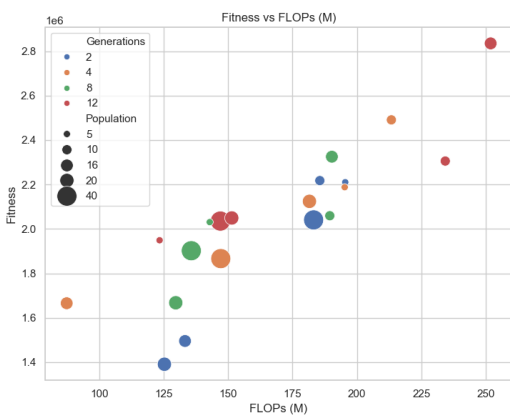
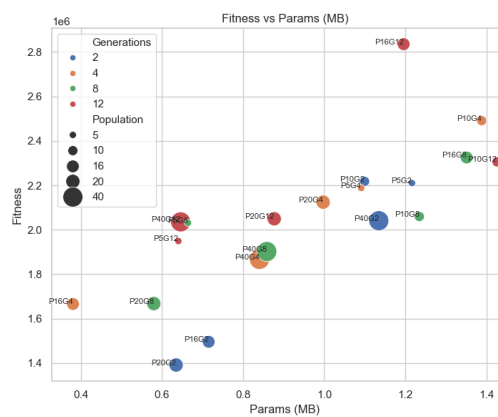
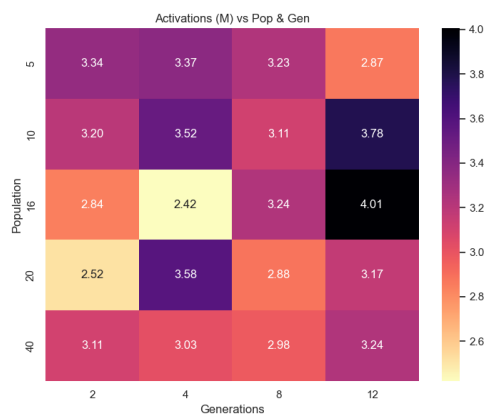
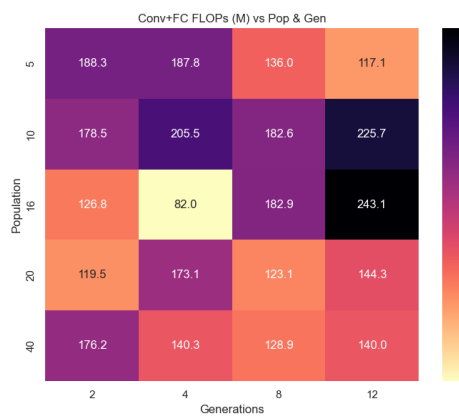
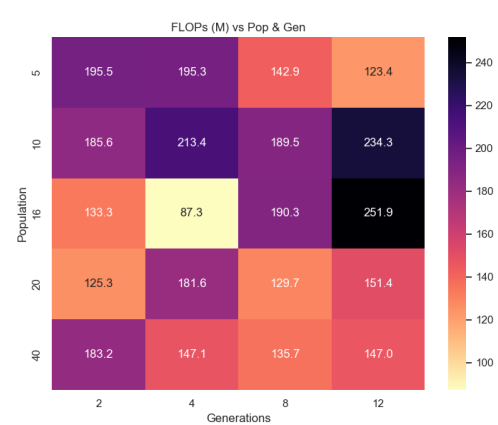
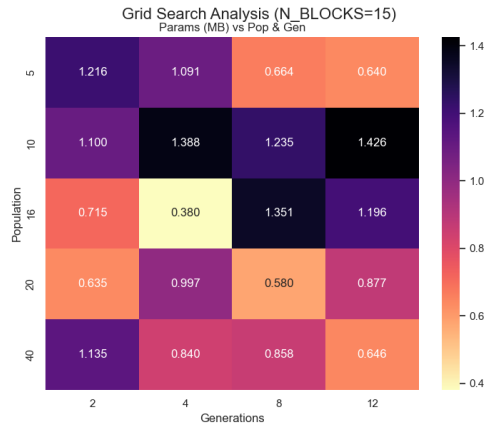
Section 2: Layer-wise Evolutionary Algorithm

Layer EA - 10 Blocks



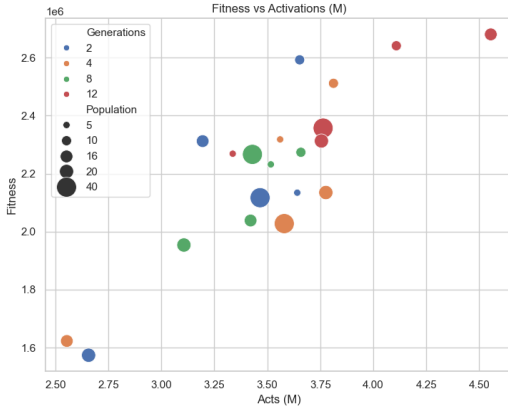
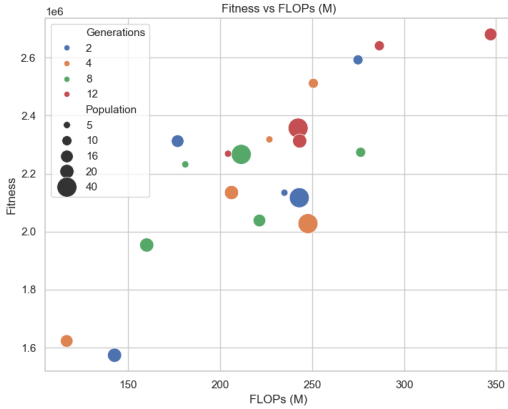
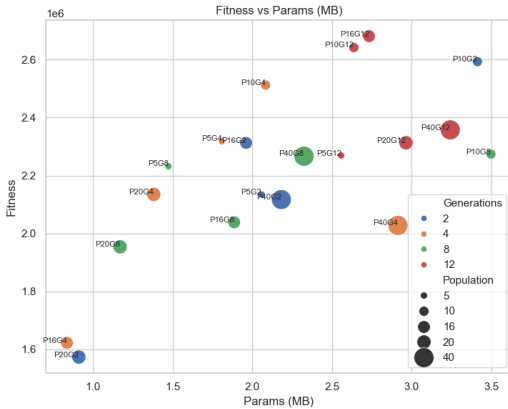
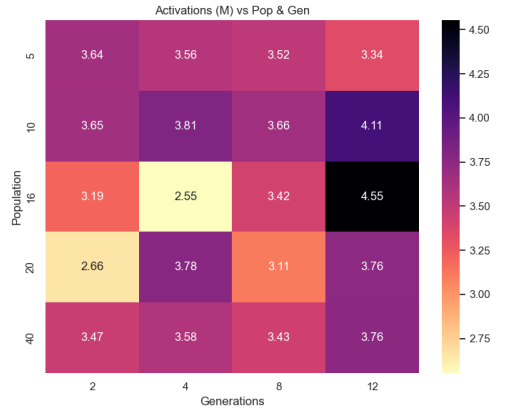
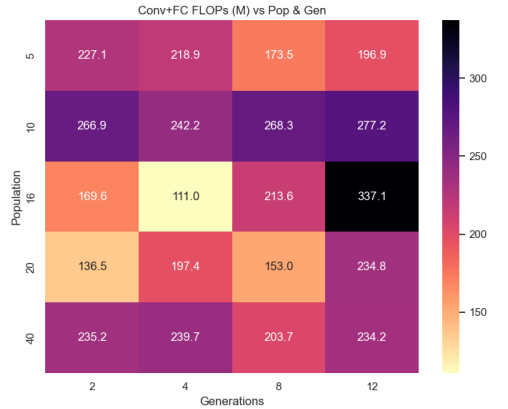
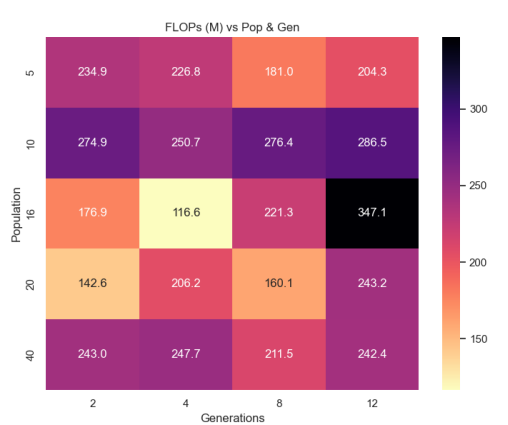
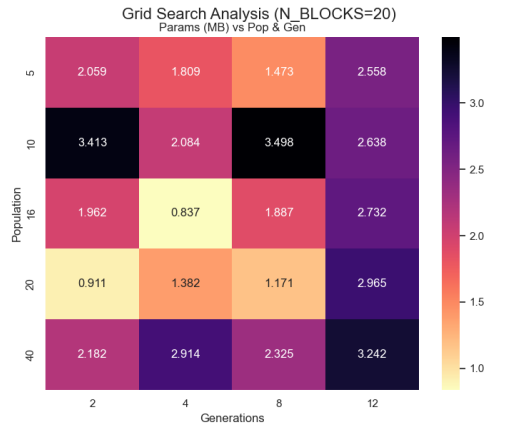
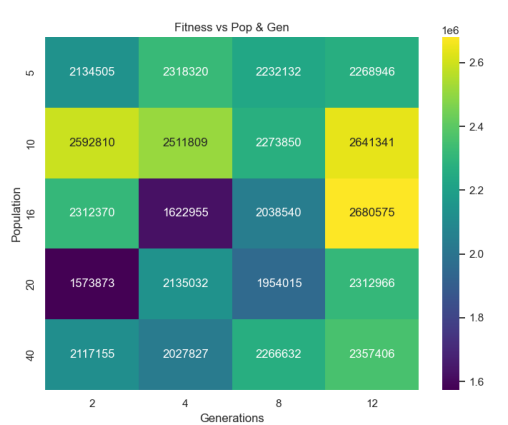
	Fitness	Params (MB)	FLOPs (M)	Conv+FC FLOPs (M)	Acts (M)
Fitness	1.0	0.4	0.53	0.53	0.71
Params (MB)	0.4	1.0	0.72	0.72	0.53
FLOPs (M)	0.53	0.72	1.0	1.0	0.87
Conv+FC FLOPs (M)	0.53	0.72	1.0	1.0	0.86
Acts (M)	0.71	0.53	0.87	0.86	1.0

Layer EA - 15 Blocks



	Fitness	Params (MB)	FLOPs (M)	Conv+FC FLOPs (M)	Acts (M)
Fitness	1.0	0.76	0.87	0.87	0.89
Params (MB)	0.76	1.0	0.91	0.92	0.74
FLOPs (M)	0.87	0.91	1.0	1.0	0.91
Conv+FC FLOPs (M)	0.87	0.92	1.0	1.0	0.9
Acts (M)	0.89	0.74	0.91	0.9	1.0

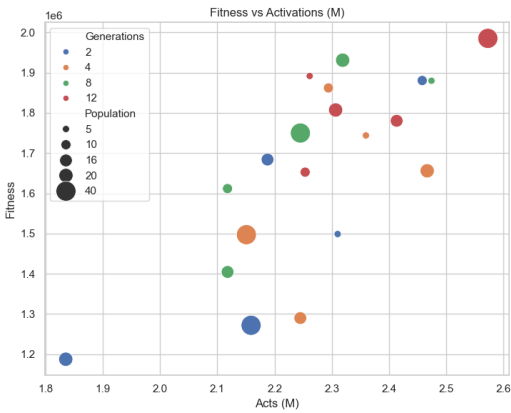
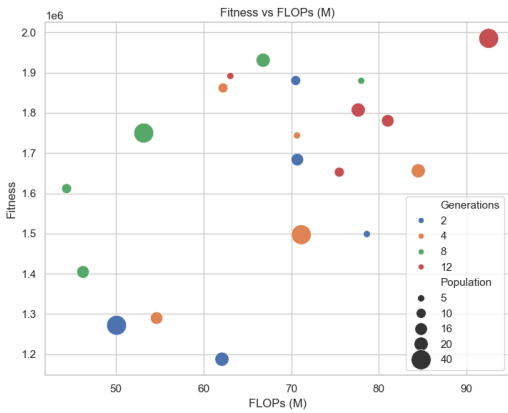
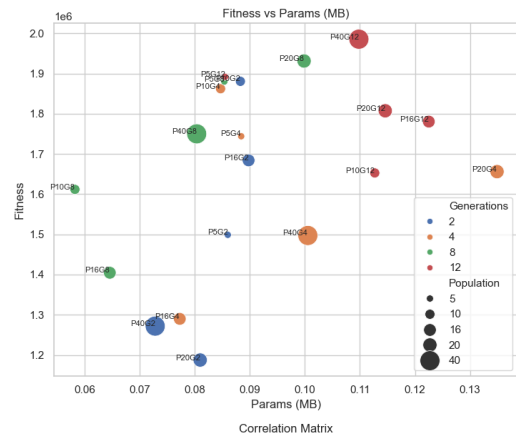
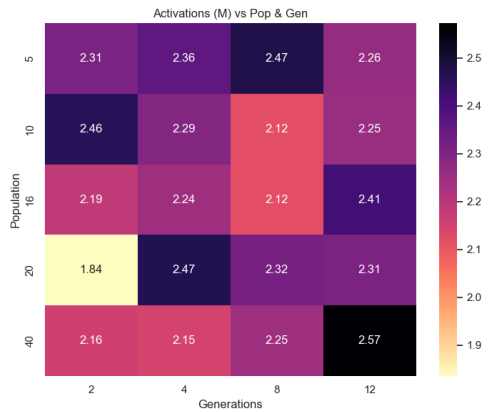
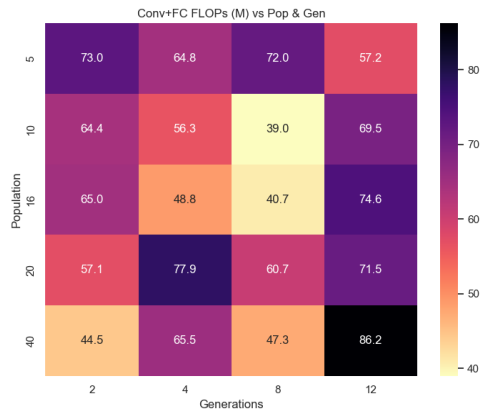
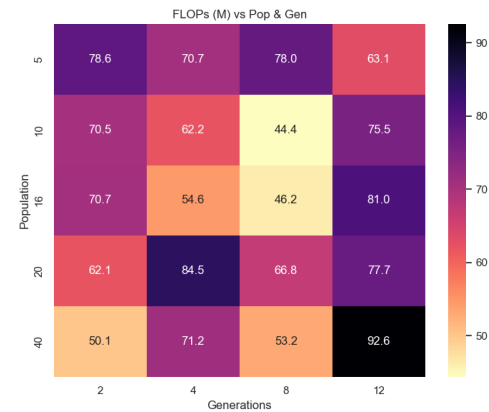
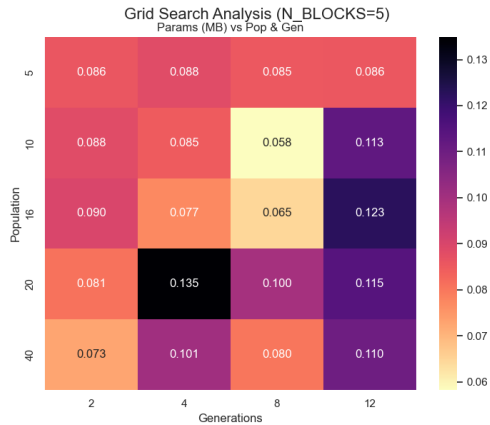
Layer EA - 20 Blocks



Correlation Matrix

	Fitness	Params (MB)	FLOPs (M)	Conv+FC FLOPs (M)	Acts (M)
Fitness	1.0	0.68	0.81	0.81	0.86
Params (MB)	0.68	1.0	0.78	0.79	0.63
FLOPs (M)	0.81	0.78	1.0	1.0	0.92
Conv+FC FLOPs (M)	0.81	0.79	1.0	1.0	0.91
Acts (M)	0.86	0.63	0.92	0.91	1.0

Layer EA - 5 Blocks



Correlation Matrix

	Fitness	Params (MB)	FLOPs (M)	Conv+FC FLOPs (M)	Acts (M)
Fitness	1.0	0.4	0.5	0.49	0.74
Params (MB)	0.4	1.0	0.83	0.83	0.54
FLOPs (M)	0.5	0.83	1.0	1.0	0.67
Conv+FC FLOPs (M)	0.49	0.83	1.0	1.0	0.66
Acts (M)	0.74	0.54	0.67	0.66	1.0